Tools of Artificial Intelligence

Evaluation metrics for classification and regression model

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Need for different evaluation metrics

- To evaluate the efficacy and effectiveness.
- Which one should I go for?
- There are plenty of ways to measure the quality of an algorithm and each developed or stakeholder decides for themselves.
- If your model is scored with some metric, you get best results by optimizing exactly that particular metric.

Example: When assessing medical images of different subjects, we need to evaluate how well a model detects correct number of healthy subjects and patients.

Types of metrics

- Classification
  - Confusion matrix
  - Accuracy
  - Precision
  - Recall (Sensitivity)
  - Specificity
  - F-1 score
  - Area under the curve and Receiver operating characteristics (AUC-ROC)
- Regression
  - Mean squared error (MSE)
  - Root MSE (RMSE)
  - Mean absolute error (MAE)
  - R-squared

Regression metrics

- Mean squared error (MSE)
- Root MSE (RMSE)
- Mean absolute error (MAE)
- R-squared

Mean square error (MSE)

Mean or average of the square of the difference between actual and estimated values.

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{1}$$

- A **high value of MSE** means that the model is not performing well
- A **low value of MSE** is desirable for well performing model
- **MSE = 0** means fits perfectly without any error.
- Why are we squaring the difference?

Root mean square error (RMSE)

RMSE is the square root of the mean of the square of all of the error.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} = \sqrt{\text{MSE}} \tag{2}$$

- RMSE **penalizes large errors** more so can be appropriate in some cases
- Advantage of RMSE over MAE is that **RMSE avoids the use of taking the absolute value**.
- RMSE provides an error metric in the same units as the target variable **(because of the square root), which makes it more intuitive**.

Mean absolute error (MAE)

MAE is the average of the absolute difference between the predicted values and observed values.

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}(|y_i - \hat{y}_i|) \tag{3}$$

- All the individual differences are weighted equally in the average.
- It doesn't tell you whether your model tends to overestimate or underestimate.
- MAE doesn't consider the direction, that is, positive or negative.

Mean absolute error (MAE)

MAE is the average of the absolute difference between the predicted values and observed values.

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}(|y_i - \hat{y}_i|) \tag{4}$$

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- MAE doesn't consider the direction, that is, positive or negative.

When to Use RMSE, MSE, and MAE

Metric	Sensitivity to Outliers	Interpretability	Use Case Examples
RMSE	High	Easy to interpret	When large errors matter (e.g., weather, finance).
MSE	High	Less interpretable	Internal tuning and optimization.
MAE	Low	Easy to interpret	When robustness to outliers is important.

Table: Comparison of RMSE, MSE, and MAE for different use cases.

Dataset for Examples

Consider the following dataset of **Actual Values** and **Predicted Values**:

Observation	**Actual Value** (y_i)	**Predicted Value** ($\hat{y}_i$)
1	3.0	2.5
2	5.0	4.0
3	2.0	2.8
4	8.0	6.5
5	6.0	7.0

Evaluate the values of MSE, RMSE, and MAE.

MAE

Formula:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

MAE

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$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Step-by-step calculation:

- Absolute Errors:

$$|3.0 - 2.5| = 0.5$$
$$|5.0 - 4.0| = 1.0$$
$$|2.0 - 2.8| = 0.8$$
$$|8.0 - 6.5| = 1.5$$
$$|6.0 - 7.0| = 1.0$$

- Sum of Absolute Errors:
  $0.5 + 1.0 + 0.8 + 1.5 + 1.0 = 4.8$
- Divide by n: $\frac{4.8}{5} = 0.96$

MAE

Formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Step-by-step calculation:

- Absolute Errors:

$$|3.0 - 2.5| = 0.5$$
$$|5.0 - 4.0| = 1.0$$
$$|2.0 - 2.8| = 0.8$$
$$|8.0 - 6.5| = 1.5$$
$$|6.0 - 7.0| = 1.0$$

- Sum of Absolute Errors:
  $0.5 + 1.0 + 0.8 + 1.5 + 1.0 = 4.8$
- Divide by n: $\frac{4.8}{5} = 0.96$

Result: The Mean Absolute Error is:

$$\text{MAE} = \mathbf{0.96}$$

MSE

Formula:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

MSE

Formula:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Step-by-step calculation:

- Squared Errors:

$$(3.0 - 2.5)^2 = 0.5^2 = 0.25$$
$$(5.0 - 4.0)^2 = 1.0^2 = 1.00$$
$$(2.0 - 2.8)^2 = (-0.8)^2 = 0.64$$
$$(8.0 - 6.5)^2 = 1.5^2 = 2.25$$
$$(6.0 - 7.0)^2 = (-1.0)^2 = 1.00$$

- Sum of Squared Errors:
  $0.25 + 1.00 + 0.64 + 2.25 + 1.00 = 5.14$
- Divide by n: $\frac{5.14}{5} = 1.028$

MSE

Formula:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Step-by-step calculation:

- Squared Errors:

$$(3.0 - 2.5)^2 = 0.5^2 = 0.25$$
$$(5.0 - 4.0)^2 = 1.0^2 = 1.00$$
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$$(8.0 - 6.5)^2 = 1.5^2 = 2.25$$
$$(6.0 - 7.0)^2 = (-1.0)^2 = 1.00$$

- Sum of Squared Errors:
  $0.25 + 1.00 + 0.64 + 2.25 + 1.00 = 5.14$
- Divide by n: $\frac{5.14}{5} = 1.028$

Result: The Mean Squared Error is:

$$\text{MSE} = 1.028$$

RMSE

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Step-by-step Calculation:

- Use the previously calculated **MSE**: 1.028.
- Take the square root: $\sqrt{1.028} \approx 1.014$.

RMSE

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Step-by-step Calculation:

- Use the previously calculated **MSE**: 1.028.
- Take the square root: $\sqrt{1.028} \approx 1.014$.

Result: The Root Mean Squared Error is:

$$\text{RMSE} \approx 1.014$$

Unsolved Example: MAE, MSE, and RMSE

Consider the following dataset for calculating the error metrics:

Observation	**Actual Value (y_i)**	**Predicted Value ($\hat{y}_i$)**
1	7.0	6.2
2	4.0	4.8
3	5.0	4.9
4	3.0	2.5
5	8.0	7.2

Task: Compute the following metrics for this dataset:

- **MAE** (Mean Absolute Error)
- **MSE** (Mean Squared Error)
- **RMSE** (Root Mean Squared Error)

Hint: Follow the same steps as in the solved example to calculate these metrics.

Answer

The computed error metrics for the unsolved example are as follows:

Metric	**Final Answer**
MAE	0.56
MSE	0.482
RMSE	0.695

R-squared (R^2) Definition

The R-squared (R^2) is a statistical measure that represents the proportion of the variance in the dependent variable that is predictable from the independent variables in a regression model.

Formula:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Where:

- y_i = Actual values (observed data points)
- $\hat{y}_i$ = Predicted values from the model
- $\bar{y}$ = Mean of the actual values
- n = Number of data points

Interpretation:

- $R^2 = 1$: Perfect fit. The model explains all the variance.
- $R^2 = 0$: The model explains none of the variance (no better than the mean).
- $R^2 < 0$: The model is worse than predicting the mean.

Classification metrics

- Confusion matrix
- Accuracy
- Precision
- Recall (Sensitivity)
- Specificity
- F-1 score
- Area under the curve and Receiver operating characteristics (AUC-ROC)

Confusion Matrix

The confusion matrix provides a detailed breakdown of classification outcomes. For a binary classification, it looks like this:

	Predicted Positive	**Predicted Negative**
Actual Positive	TP(True Positive)	FN(False Negative)
Actual Negative	FP(False Positive)	TN(True Negative)

Definitions:

- **True Positive (TP):** Correctly predicted positive cases.
- **True Negative (TN):** Correctly predicted negative cases.
- **False Positive (FP):** Negative cases incorrectly predicted as positive.
- **False Negative (FN):** Positive cases incorrectly predicted as negative.

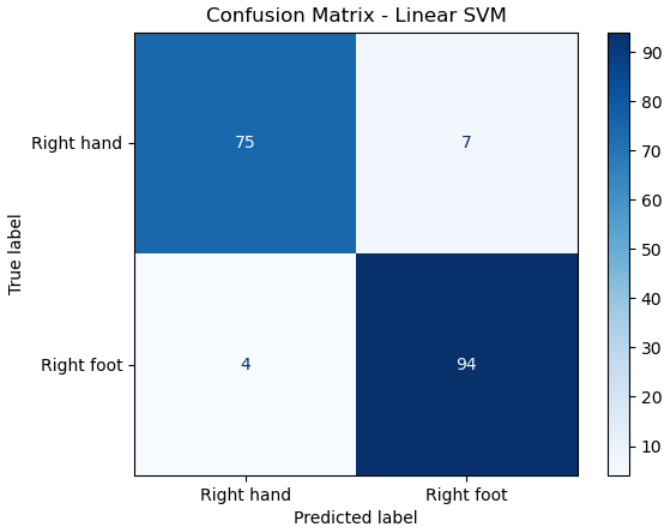
Confusion matrix example and usage

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
# Plot confusion matrices for both models
y_pred_linear = linear_svm.predict(X_test)
y_pred_nonlinear = nonlinear_svm.predict(X_test)

cm_linear = confusion_matrix(y_test, y_pred_linear)
cm_nonlinear = confusion_matrix(y_test, y_pred_nonlinear)

disp_linear = ConfusionMatrixDisplay(confusion_matrix=cm_linear,
display_labels=['Right hand', 'Right foot'])
disp_nonlinear = ConfusionMatrixDisplay(confusion_matrix=cm_nonlinear,
display_labels=['Right hand', 'Right foot'])

disp_linear.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix - Linear SVM")
plt.savefig("CM_L_SVM")
plt.show()
```



Accuracy

Accuracy measures the overall correctness of the model, representing the proportion of correct predictions out of all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

When to use:

- Useful when the dataset is balanced (equal proportion of positive and negative cases).
- Not ideal for imbalanced datasets where one class dominates.

Precision

Precision (Positive Predictive Value) measures the proportion of correctly identified positive cases out of all cases predicted as positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

When to use:

- Useful when the cost of false positives is high.
- Example: Cancer diagnosis where unnecessary treatment can have severe consequences.

Recall (Sensitivity)

Recall (or Sensitivity) measures the proportion of actual positive cases that are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

When to use:

- Useful when the cost of false negatives is high.
- Example: Early-stage cancer screening to minimize missed diagnoses.

Specificity

Specificity (True Negative Rate) measures the proportion of actual negative cases that are correctly identified by the model.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

When to use:

- Useful when the cost of false positives is high.
- Example: Diagnosing non-cancer cases in a population to minimize false alarms.

F-1 Score

The **F-1 Score** is the harmonic mean of Precision and Recall, providing a single metric that balances both.

$$F-1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

When to use:

- Useful when there is an imbalance between classes and both Precision and Recall are important.
- Example: Fraud detection in financial transactions where both false positives and false negatives need attention.

Performance report example and usage

```python
from sklearn.metrics import classification_report
report_linear = classification_report(y_test, y_pred_linear,
      target_names=['Right hand', 'Right foot'])
report_nonlinear = classification_report(y_test, y_pred_nonlinear,
      target_names=['Right hand', 'Right foot'])
print(report_linear)
print(report_nonlinear)
```

```
              precision    recall  f1-score   support

  Right hand       0.95      0.91      0.93        82
  Right foot       0.93      0.96      0.94        98

    accuracy                           0.94       180
   macro avg       0.94      0.94      0.94       180
weighted avg       0.94      0.94      0.94       180
```

AUC-ROC: Area Under the Curve - Receiver Operating Characteristics

Definition:

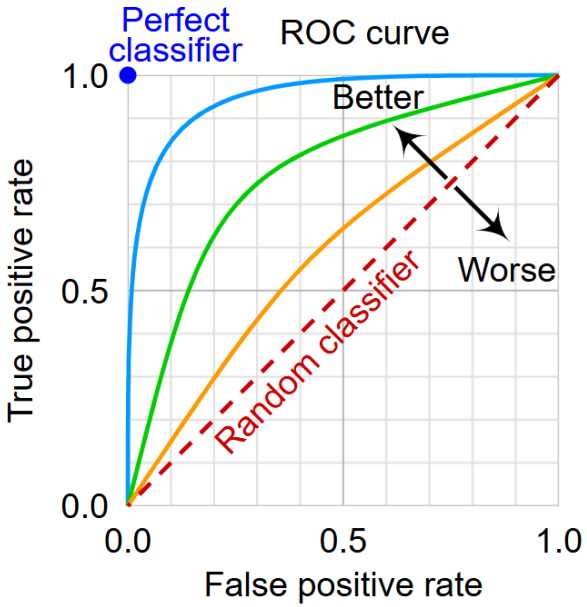
- **AUC (Area Under the Curve):** Measures the ability of a classifier to distinguish between positive and negative classes across all thresholds.
- **ROC (Receiver Operating Characteristics):** A curve plotting the True Positive Rate (TPR or Recall) against the False Positive Rate (FPR).

Key Points:

- AUC ranges from 0 to 1:
  - AUC $= 1$: Perfect classifier.
  - AUC $= 0.5$: Random guess (no discrimination).
  - AUC < 0.5: Worse than random guessing.
- Higher AUC indicates better model performance.

AUC-ROC

Graphical Representation:



Interpretation:

- The closer the curve is to the top-left corner, the better the performance.
- AUC represents the probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance.

Source: By cmglee, MartinThoma — Roc-draft-xkcd-style.svg, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=109730045

ROC AUC example and usage

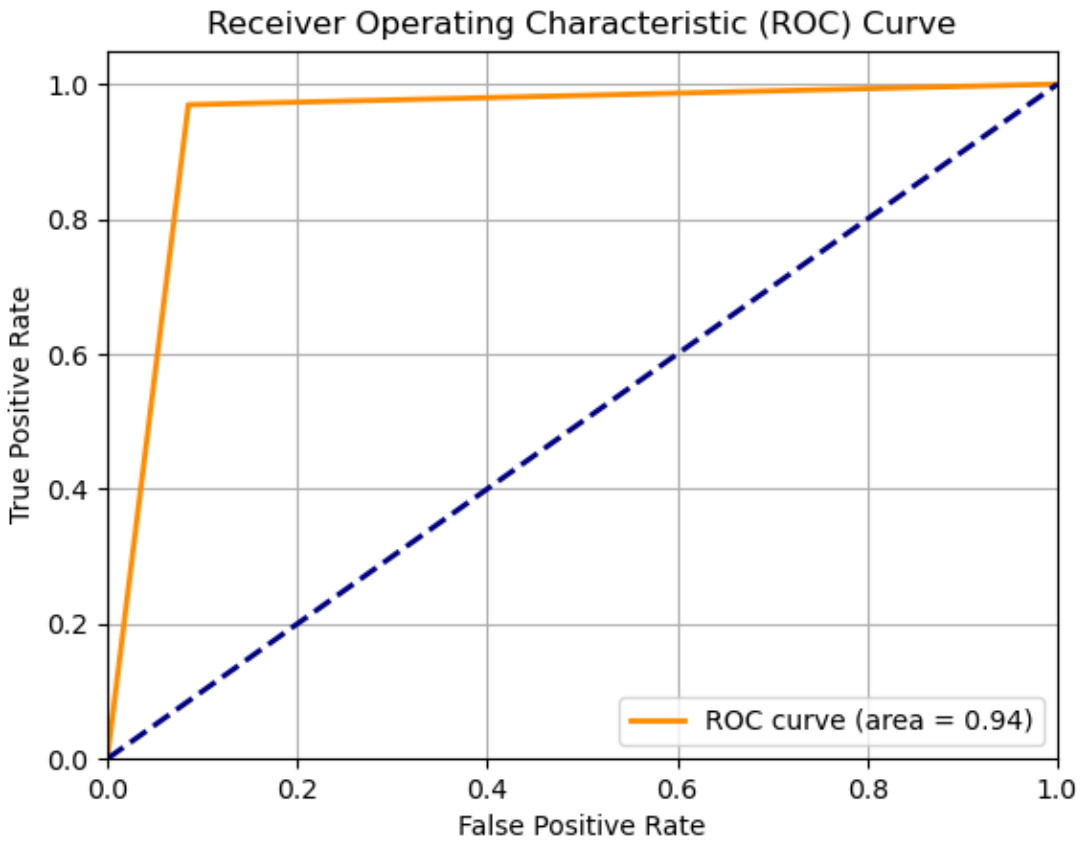
```
from sklearn.metrics import roc_curve, auc

# Compute ROC curve and ROC area
fpr, tpr, thresholds_roc = roc_curve(y_test, y_pred_nonlinear)
roc_auc = auc(fpr, tpr)

# Compute Precision-Recall curve and average precision score
precision, recall, thresholds_pr = precision_recall_curve(y_test,
    y_pred_nonlinear)
average_precision = average_precision_score(y_test, y_pred_nonlinear)

# Plot ROC curve with detailed annotations
plt.figure()
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area = {
    roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc="lower right")
plt.grid(True)
plt.savefig('ROC')
```

ROC AUC report example



When to Use Which Metric?

Metric	Best Use Case	Example Scenario
Accuracy	When the dataset is balanced and overall performance matters.	Diagnosing common conditions with equal prevalence of positive and negative cases.
Precision	When false positives must be minimized (e.g., cost of a false alarm is high).	Cancer detection where unnecessary treatments have severe consequences.
Recall (Sensitivity)	When false negatives must be minimized (e.g., cost of missing a true case is high).	Screening for cancer to ensure no cases are missed.
Specificity	When false positives must be minimized and you need to confirm true negatives.	Confirming non-cancer cases in a secondary diagnostic stage.
F-1 Score	When you need a balance between Precision and Recall, especially for imbalanced datasets.	Fraud detection in financial transactions.
AUC-ROC	When you want to evaluate overall model performance across thresholds.	Comparing models for medical diagnosis where sensitivity and specificity vary with the threshold.

Table: Summary of Metrics and Their Best Use Cases

Solved Example: Cancer Detection Model

Confusion Matrix:

	Predicted Positive	**Predicted Negative**
Actual Positive	90 (TP)	10 (FN)
Actual Negative	15 (FP)	85 (TN)

Solved Example: Cancer Detection Model

Metrics Calculation:

Metric	Formula	Value
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	$\frac{90+85}{90+85+15+10} = 0.875$
Precision	$\frac{TP}{TP+FP}$	$\frac{90}{90+15} = 0.857$
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	$\frac{90}{90+10} = 0.9$
Specificity	$\frac{TN}{TN+FP}$	$\frac{85}{85+15} = 0.85$
F1 Score	$\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}}$	$\frac{2 \cdot 0.857 \cdot 0.9}{0.857+0.9} \approx 0.878$

Table: Summary of Metrics and Calculations

Unsolved Example: Cancer Detection Model

Confusion Matrix:

	Predicted Positive	**Predicted Negative**
Actual Positive	100 (TP)	20 (FN)
Actual Negative	10 (FP)	120 (TN)

Tasks:

1. Calculate the **Accuracy**.
2. Calculate the **Precision**.
3. Calculate the **Recall (Sensitivity)**.
4. Calculate the **Specificity**.
5. Calculate the **F1 Score**.

Answer

Metric	Value
Accuracy	$\frac{100+120}{100+120+10+20} = 0.92$
Recall (Sensitivity)	$\frac{100}{100+20} = 0.833$
Specificity	$\frac{120}{120+10} = 0.923$
Precision	$\frac{100}{100+10} = 0.909$
F1 Score	$\frac{2 \cdot 0.909 \cdot 0.833}{0.909+0.833} = 0.87$

Table: Summary of Metrics